Database Systems and Information Storage Architectures

Academic Lecture Notes: WSM Information Retrieval Foundations

1. Overview of Database Management Systems (DBMS)
A database management system is software designed to store, manage, and facilitate structured access
to large quantities of data. Relational database systems utilize SQL for deterministic querying
and maintain ACID (Atomicity, Consistency, Isolation, Durability) properties.

2. Comparison: Database Systems vs. Information Retrieval Systems
While traditional database systems excel at exact transactional record queries (e.g. employee ID lookup),
Information Retrieval (IR) engines handle unstructured text documents. In an IR system, relevance scoring,
term frequency, and wildcard pattern expansion take precedence over tabular schemas.

3. Indexing Mechanisms: B-Trees, Hash Tables, and Inverted Lists
Database storage architectures leverage B-Tree and B+ Tree indexes for fast range queries over ordered keys.
In contrast, search engines use inverted index postings lists, which map individual vocabulary tokens to document identifiers and word positions.

4. Query Processing and Optimization
Database engines translate declarative SQL queries into physical relational algebra plans. Similarly, retrieval engines parse multi-term queries and evaluate wildcard expressions like 'data*base'
to efficiently retrieve all matching records.